

# Ajay Suthaharan

647-554-9642 | [asuth075@uottawa.ca](mailto:asuth075@uottawa.ca) | [linkedin.com/in/ajaysuthaharan/](https://www.linkedin.com/in/ajaysuthaharan/) | [github.com/Ajay757](https://github.com/Ajay757)

## EDUCATION

### University of Ottawa

Ottawa, ON

Honours Bachelor of Science in Computer Science (Data Science Option) GPA: 3.8/4

Sep. 2023 – May. 2028

**Relevant Coursework:** Data Structures & Algorithms, Operating Systems, Software Engineering, Databases I & II, Computer Architecture, Data Communication & Networking

## TECHNICAL SKILLS

**Languages:** Java, Python, C#, JavaScript, SQL, HTML/CSS, Ruby, R

**Frameworks & Libraries:** React, Laravel, TensorFlow/Keras, Pandas, NumPy, FastAPI, pgvector

**Databases:** PostgreSQL, MySQL, Firebase, MariaDB, pgvector

**Developer Tools:** Git, Gitlab, Docker, VS Code, Google Colab, Codex, Claude, XCode

## EXPERIENCE

### Statistical Programmer Co-op

Sep. 2026 – Present

*Labour Force Survey, Statistics Canada*

*Toronto, ON, Canada*

- Responsible for SAS-to-R migration of LFS workflows calculating standard errors for seasonally adjusted estimates
- Tasked with developing R pipelines for provincial and territorial bootstrap samples and survey weights.
- Validation responsibilities include comparing R outputs against SAS results, documenting workflows, and managing code through GitLab.

### Machine Learning Research Engineer Co-op

Jan. 2026 – Apr. 2026

*Mapping and Earth Observation, Natural Resources Canada*

*Ottawa, ON, Canada*

- Evaluated vision, vision-language, and spatiotemporal foundation models for Earth observation applications.
- Developed a taxonomy classifying models by spatial, temporal, spectral, SAR, and cross-modal characteristics.
- Built structured comparison tables to standardize evaluation across emerging geospatial AI models.

### Software Developer Internship

May. 2024 – Aug. 2024

*Driftstack*

*Greater Toronto, ON*

- Revamped and optimized the website UI with React, improving navigation and reducing bounce rates
- Contributed to the development of RESTful APIs, enabling seamless third-party platform connections.
- Implemented AppSumo integration workflows and real-time data synchronization using Laravel, MySQL, and MariaDB.

## PROJECTS

### SkySense – Flight Analytics Dashboard | *Python, React, JavaScript, Node.js*

Jan. 2026

- Built an airspace simulation platform supporting 250–1000 simulated flights with multi-aircraft conflict detection.
- Implemented trajectory simulation and conflict-detection logic to identify loss-of-separation events.
- Integrated a Python backend with a React dashboard through a custom API to visualize flight paths, conflicts, and resolution status.

### HiPREX – AI Agent Backend | *Python, FastAPI, PostgreSQL, pgvector, Docker*

Jan. 2025 – Present

- Built FastAPI request routing that separates LLM planning from deterministic execution, memory lookup, and state updates.
- Implemented structured update actions, nested state mutation, and session/message persistence to retain information across conversations.
- Built PostgreSQL/pgvector infrastructure for embeddings and similarity retrieval, and exposed routing and memory results as API response metadata.

### e-Hotels – Hotel Booking Application | *Java, PostgreSQL, JDBC, JavaScript*

Jan. 2025 – Apr. 2025

- Designed a relational database schema for hotels, customers, employees, and bookings.
- Developed Java backend services for room search, reservations, and administrative CRUD operations, connected to a web frontend.
- Implemented SQL queries and joins to filter available rooms by capacity, price, city, and hotel chain.